

## SAFETY DATA SHEET

Creation Date 28-October-2009

Revision Date 22-April-2024

Revision Number 1

### 1. Identification

**Product Name** Hydrogen Peroxide, 28%, Extra Pure

**Cat No. :** S60390

**Synonyms** Hydrogen Dioxide; Peroxide; Carbamide Peroxide

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |              |
|------------------------------------------|--------------|
| <b>Oxidizing liquids</b>                 | Category 2   |
| <b>Acute oral toxicity</b>               | Category 4   |
| <b>Acute Inhalation Toxicity</b>         | Category 4   |
| <b>Skin Corrosion/Irritation</b>         | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 1   |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May intensify fire; oxidizer

Harmful if swallowed or if inhaled

Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep/Store away from clothing/combustible materials

Take any precaution to avoid mixing with combustibles

Avoid breathing dust/fume/gas/mist/vapors/spray

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

Use only outdoors or in a well-ventilated area

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Rinse mouth

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Light sensitive

### 3. Composition/Information on Ingredients

| Component         | CAS-No    | Weight % |
|-------------------|-----------|----------|
| Water             | 7732-18-5 | 65 - 80  |
| Hydrogen peroxide | 7722-84-1 | 20 - 35  |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms/effects**

None reasonably foreseeable. Causes eye burns.

**Notes to Physician**

Treat symptomatically

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Use water spray or fog; do not use straight streams.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** 100%

**Lower** 40%

**Oxidizing Properties** Oxidizer

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Corrosive material. Containers may explode when heated. Oxidizer: Contact with combustible/organic material may cause fire. In the event of fire and/or explosion do not breathe fumes. Thermal decomposition can lead to release of irritating gases and vapors. May ignite combustibles (wood paper, oil, clothing, etc.).

**Hazardous Combustion Products**

Hydrogen. Oxygen.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
OX

## 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required.

Do not use steel or aluminum tools or equipment

**Environmental Precautions** Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. To maintain product quality. Keep refrigerated. Protect from direct sunlight. Do not store in metal containers. Containers should be vented periodically in order to overcome pressure buildup. Do not store near combustible materials. Incompatible Materials. Strong oxidizing agents. Metals. Reducing Agent. Alcohols. Ammonia. copper. Copper alloys. lead oxides. Cyanides. Sulfides. Lead. Acetone. Aluminium. . Strong reducing agents. Combustible material.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component         | Alberta                                  | British Columbia | Ontario TWAEV | Quebec     | ACGIH TLV  | OSHA PEL             | NIOSH                      |
|-------------------|------------------------------------------|------------------|---------------|------------|------------|----------------------|----------------------------|
| Hydrogen peroxide | TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup> | TWA: 1 ppm       | TWA: 1 ppm    | TWA: 1 ppm | TWA: 1 ppm | (Vacated) TWA: 1 ppm | IDLH: 75 ppm<br>TWA: 1 ppm |

|  |  |  |  |  |  |                                                                                     |                            |
|--|--|--|--|--|--|-------------------------------------------------------------------------------------|----------------------------|
|  |  |  |  |  |  | (Vacated) TWA:<br>1.4 mg/m <sup>3</sup><br>TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup> | TWA: 1.4 mg/m <sup>3</sup> |
|--|--|--|--|--|--|-------------------------------------------------------------------------------------|----------------------------|

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Butyl rubber   | > 480 minutes     | 0.35 mm         | Splash protection only |
| Neoprene       | > 480 minutes     | 0.45 mm         |                        |
| Natural rubber | > 480 minutes     | 0.5 mm          |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Inorganic gases and vapours filter Type B Grey conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                          |                              |
|--------------------------|------------------------------|
| Physical State           | Liquid                       |
| Appearance               | Colorless                    |
| Odor                     | Slight                       |
| Odor Threshold           | No information available     |
| pH                       | 3.3                          |
| Melting Point/Range      | -33 °C / -27.4 °F            |
| Boiling Point/Range      | 108 °C / 226.4 °F @ 760 mmHg |
| Flash Point              | No information available     |
| Evaporation Rate         | 1.0 (Butyl acetate = 1.0)    |
| Flammability (solid,gas) | Not applicable               |

**Flammability or explosive limits**

|                                        |                          |
|----------------------------------------|--------------------------|
| Upper                                  | 100%                     |
| Lower                                  | 40%                      |
| Vapor Pressure                         | No information available |
| Vapor Density                          | 1.10                     |
| Specific Gravity                       | 1.110                    |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | > 125°C                  |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                         |                                                                                                                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                                                                                                                  |
| <b>Stability</b>                        | Sensitivity to light. Oxidizer: Contact with combustible/organic material may cause fire.                                                                                                            |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Exposure to light. Combustible material.                                                                                                                         |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Metals, Reducing Agent, Alcohols, Ammonia, copper, Copper alloys, lead oxides, Cyanides, Sulfides, Lead, Acetone, Aluminium, , Strong reducing agents, Combustible material |
| <b>Hazardous Decomposition Products</b> | Hydrogen, Oxygen                                                                                                                                                                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                                             |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                                                                        |

## 11. Toxicological information

**Acute Toxicity****Product Information**

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| <b>Oral LD50</b>   | Category 4. ATE = 300 - 2000 mg/kg.                                           |
| <b>Dermal LD50</b> | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Mist LC50</b>   | Category 4. ATE = 1 - 5 mg/l.                                                 |

**Component Information**

| Component         | LD50 Oral                                                                               | LD50 Dermal            | LC50 Inhalation                           |
|-------------------|-----------------------------------------------------------------------------------------|------------------------|-------------------------------------------|
| Water             | -                                                                                       | -                      | -                                         |
| Hydrogen peroxide | 376 mg/kg ( Rat ) (90%)<br>910 mg/kg ( Rat ) (20-60%)<br>1518 mg/kg ( Rat ) (8-20% sol) | >2000 mg/kg ( Rabbit ) | LC50 = 2000 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic** No information available**Products****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | CAUSES (SEVERE) EYE BURNS May cause irritation                                           |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component         | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|-----------|------------|------------|------------|------------|------------|
| Water             | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Hydrogen peroxide | 7722-84-1 | Not listed | Not listed | A3         | Not listed | A3         |

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens

A1 - Known Human Carcinogen  
A2 - Suspected Human Carcinogen  
A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)  
Mexico - Occupational Exposure Limits - Carcinogens

A1 - Confirmed Human Carcinogen  
A2 - Suspected Human Carcinogen  
A3 - Confirmed Animal Carcinogen  
A4 - Not Classifiable as a Human Carcinogen  
A5 - Not Suspected as a Human Carcinogen

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component         | Freshwater Algae  | Freshwater Fish                     | Microtox   | Water Flea        |
|-------------------|-------------------|-------------------------------------|------------|-------------------|
| Hydrogen peroxide | EC50 2.5 mg/L/72h | LC50: 16.4 mg/L/96h<br>(P.promelas) | Not listed | EC50 7.7 mg/L/24h |

**Persistence and Degradability** Persistence is unlikely Decomposes Soluble in water based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component         | log Pow |
|-------------------|---------|
| Hydrogen peroxide | -1.1    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2014  
Proper Shipping Name HYDROGEN PEROXIDE, AQUEOUS SOLUTIONS  
Hazard Class 5.1  
Subsidiary Hazard Class 8  
Packing Group II

### TDG

UN-No UN2014

|                         |                                      |
|-------------------------|--------------------------------------|
| Proper Shipping Name    | HYDROGEN PEROXIDE, AQUEOUS SOLUTIONS |
| Hazard Class            | 5.1                                  |
| Subsidiary Hazard Class | 8                                    |
| Packing Group           | II                                   |

**IATA**

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2014                              |
| Proper Shipping Name    | HYDROGEN PEROXIDE, AQUEOUS SOLUTION |
| Hazard Class            | 5.1                                 |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | II                                  |

**IMDG/IMO**

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2014                              |
| Proper Shipping Name    | HYDROGEN PEROXIDE, AQUEOUS SOLUTION |
| Hazard Class            | 5.1                                 |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | II                                  |

## 15. Regulatory information

**International Inventories**

| Component         | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Water             | 7732-18-5 | X   | -    | X    | ACTIVE                                        | 231-791-2 | -      | -   |
| Hydrogen peroxide | 7722-84-1 | X   | -    | X    | ACTIVE                                        | 231-765-0 | -      | -   |

| Component         | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Water             | 7732-18-5 | X     | KE-35400 | X    | -    | X    | X    | X     | X     |
| Hydrogen peroxide | 7722-84-1 | X     | KE-20204 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrogen peroxide | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

**REACH links**
<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component         | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water             | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Hydrogen peroxide | 7722-84-1 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component         | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water             | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Hydrogen peroxide | 7722-84-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 28-October-2009  
**Revision Date** 22-April-2024  
**Print Date** 22-April-2024  
**Revision Summary** Initial Release.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**